

ALGEBRA REVIEW

EXPONENTS

Definition (Positive Integer Exponents). Given a real number x and positive integer n , we define

$$x^n = x \cdot x \cdot x$$

where there are n factors of x in this product. Moreover, if $x \neq 0$, then we take $x^0 = 1$ and note 0^0 is not defined.

Definition (Negative Integer Exponents). Given a real number x and a positive integer n , we define

$$x^{-n} = \frac{1}{x^n}.$$

Example 1. 1. $3^3 = 3 \cdot 3 \cdot 3 = 27$

2. $\left(\frac{1}{\pi}\right)^0 = 1$

3. $4^{-2} = \frac{1}{4^2} = \frac{1}{16}$

4. $\left(\frac{1}{5}\right)^2 = \frac{1}{5} \frac{1}{5} = \frac{1}{25}.$

△

Theorem 1. Let m and n be positive integers and x a real number. Then

1. $x^m x^n = x^{m+n}$

2. $(x^m)^n = x^{mn}$

3. $(xy)^n = x^n y^n$

4. $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$

5. $\frac{x^n}{x^m} = x^{n-m}$ if $x \neq 0$

6. $\frac{x^n}{x^m} = \frac{1}{x^{m-n}}$ if $x \neq 0$

Example 2. Simplify $\left(\frac{x^{-2}y^2}{x^{-4}y^5}\right)$ and express the answer using no negative exponents.

△

ROOTS

Definition (Roots). Suppose that for some positive integer n , $y^n = x$. Then we say that y is an n th root of x .

Example 3. 1. Since $2^2 = 4$, we say that 2 is a second root of 4.

2. Since $3^3 = 27$, we say that 3 is a third root of 27.

△

Definition.

1. If n is odd, we write $\sqrt[n]{x} = x^{1/n}$ and note there is only one n th root.
2. If n is even and x is odd, there are no real n th roots of x (there are no real roots of -1 since $x^2 = -1$ has no solutions).
3. If n is even and x is nonnegative, we write $\sqrt[n]{x} = x^{1/n}$ to mean the positive root of x (note there are two n th roots in this case).
4. If $n = 2$, then we write $\sqrt{2x} = \sqrt{x}$.

Theorem 2. Let n be a positive integer and x and y take on values that make sense. Then

1. $\sqrt[n]{x}y = \sqrt[n]{x}\sqrt[n]{y} = (xy)^{1/n} = x^{1/n}y^{1/n}$
2. $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}} = \left(\frac{x}{y}\right)^{1/n} = \frac{x^{1/n}}{y^{1/n}}$
3. $\sqrt[n]{x}\sqrt[m]{x} = \sqrt[nm]{x} = (x^{1/m})^{1/n} = x^{1/mn}$

Definition (Rational Exponents). If m and n are integers, then

$$x^{m/n} = (x^{1/n})^m = (x^m)^{1/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m},$$

so long as x makes sense.

Example 4. Rewrite $\sqrt[4]{\frac{2x^{16}}{y^{-7}}}$ in the most simplified manner using no radicals or negative exponents. $\triangle$

POLYNOMIALS

Definition (Polynomials). Let n be a nonnegative integer, $a_0, a_1, \dots, a_n$ real numbers with at least $a_n \neq 0$, then we call

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

a polynomial.

Example 5. 1. $1 + x + x^2$

2. $x^5 + x + \pi$.

$\triangle$

Definition (Common polynomials). 1. Polynomials of the form a_0 are called constant polynomials.

2. Polynomials of the form $a_1 x + a_0$ are called linear polynomials.

3. Polynomials of the form $a_2 x^2 + a_1 x + a_0$ are called quadratic polynomials.

4. Polynomials of the form $a_3 x^3 + a_2 x^2 + a_1 x + a_0$ are called cubic polynomials.

Example 6. Simplify the following polynomial expressions.

1. $(x - 1)(x + 1)$

2. $(ax^2 + bx + c) - (x^2 + dx + e)$

3. $(2x + 2)(x^3 + 2x + 1)$.

$\triangle$

FACTORING POLYNOMIALS

Theorem 3. Let $p(x) = ax^2 + bx + c$. Then the x such that $p(x) = 0$ is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Example 7. 1. Factor $x^2 + 2x + 1$

2. Factor $x^2 + 4x + 3$

3. Factor $x^3 + 2x^2 + x$

4. Factor $2x^2 + 8x + 3$

△

Theorem. For a quadratic $ax^2 + bx + c$, the equation $ax^2 + bx + c = 0$ has a real solution if and only if

$$b^2 - 4ac \geq 0.$$

Example 8. Consider $x^2 + 1$. Here $a = 1$, $b = 0$, and $c = 1$. Then $0^2 - 4(1)(1) = -4 < 0$ and so $x^2 + 1 = 0$ has no solutions. △